

Autonomous book retrieval on TIAGo Pro

Emirates Robotics Competition 2026

Phase 1 — Simulation

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Repository

github.com/4waiz/Rafeeq-library

Validated against

organiser dfl-rlab/erc_sim_2026 @ 2aa5a4a (v1.0.3-1-g2aa5a4a)

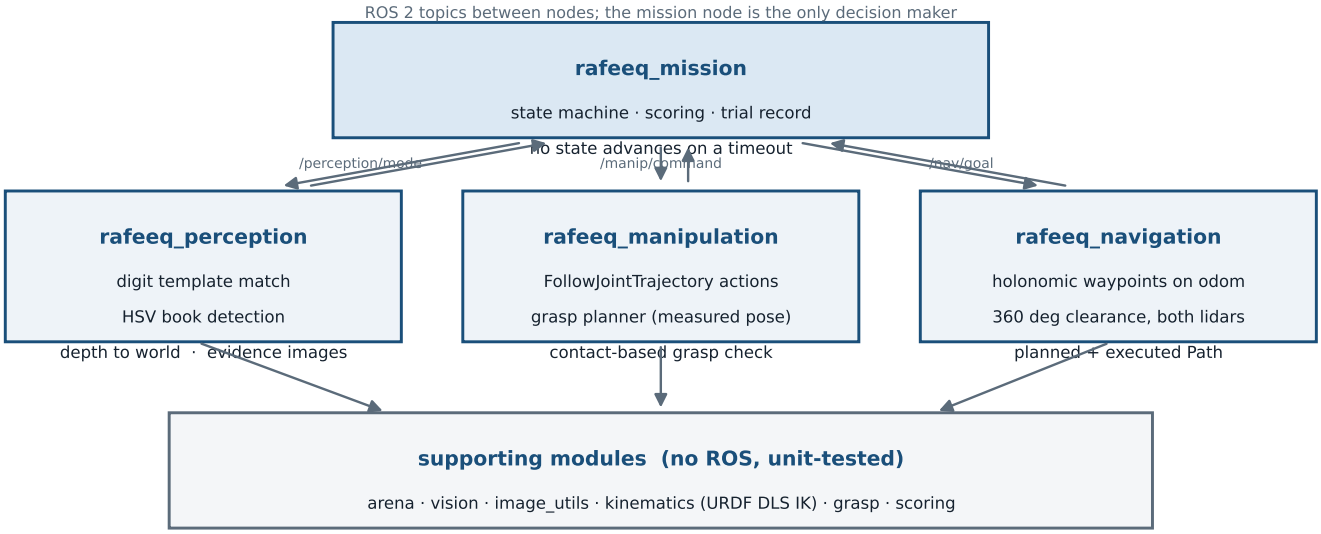
Ubuntu 22.04 · ROS 2 Humble · Gazebo Sim 8.15.0 (Harmonic)

Task completed end to end on the live simulator: book_col_3_row_2_red in the collection bin at t+107.8 s, reported by the organisers' own /bin_contacts sensor. One unedited take, 4:34.

That is one run, not a rate: the carry is timing-critical and runs still drop the book (section 5).

1 System architecture

Rafeeq is four ROS 2 nodes plus six supporting modules that carry no ROS dependency and are unit-tested offline. The mission node is the only component that decides anything: perception, navigation and manipulation each expose a narrow command-and-status contract and never talk to each other. That separation is what made the defects in this system findable at all.



Node responsibilities

Node	Responsibility	Key interfaces
rafeeq_mission	State machine, scoring, trial record	/rafeeq/*, /bin_contacts, /contacts
rafeeq_perception	Markers, books, bin, evidence images	RGB-D, CameraInfo, TF, /erc/*
rafeeq_navigation	Holonomic waypoints, clearance	/cmd_vel, /odom, /scan_*_raw
rafeeq_manipulation	Arm, torso, head, gripper	FollowJointTrajectory actions, /joint_states

Perception to action, end to end

The mission asks perception for a column; perception reads the number plates, deprojects each into the world, and publishes the physical column together with an annotated image. The mission drives there and asks for the book; perception returns the row and, critically, the measured 3D position. The mission passes that point to manipulation, which plans a grasp posture against it, and drives the base so the book is squarely in front of the arm. Nothing in this chain uses a nominal column-centre position: the organiser jitters each book by up to 0.25 m, so a nominal target misses by three times the gripper opening.

What is deliberately not used

- Nav2 is installed but never brought up by the organiser launch: no map server, no AMCL, no costmap, no map frame. Rafeeq drives odometry waypoints and publishes the intended waypoint path and the executed trajectory as nav_msgs/Path, labelled as such rather than as a Nav2 plan.
- MoveIt is installed but has nothing to plan against: the repository ships no SRDF, planning group or kinematics plugin for TIAGo Pro. kinematics.py parses the competition URDF and solves the 8-DOF chain with damped least squares.

2 Perception

Shelf number identification

The number plates are a black digit on a white square, and the exact digit textures ship inside erc_description. That makes template matching against the ground-truth textures strictly better than OCR or a learned detector here: five classes, no dataset, no training, and no failure mode that is not visible in the mask. Dark ink is segmented in HSV, each blob is tight-cropped, normalised to 48x64 and scored by intersection-over-union against every template. A digit is accepted at IoU 0.55 and must agree on three consecutive frames before it is published.

Mapping a digit to a physical column is done by measurement, not assumption: each plate is deprojected through the depth image into the world and snapped to the nearest column centre. Plates, unlike books, are not jittered - measured at exactly y = +2.0, +1.0, 0.0, -1.0, -2.0. If depth or TF is unavailable for some plates and all five are in frame, their left-to-right image order is used instead, and the fallback is logged.

Book colour, row and position

Books are unlit, fully saturated primaries presented spine-on as a 0.03 x 0.25 m rectangle, so a tight HSV band plus a 1:8 aspect test rejects almost every false positive. The row comes from two independent methods that cross-check each other: the vertical order of the four books in a column, and the measured world height snapped to the nearest row. A disagreement is reported rather than silently resolved.

The output that matters is the measured 3D position. If depth or TF is unavailable the mode fails rather than publishing a column centre - a row number alone cannot be grasped. The column filter refuses to fall back to "all books in the image": every column holds all four colours, so such a fallback would happily return a same-colour book a metre away in a neighbouring column.

[missing image: erc_images\shelf_column_20260909_100140_814.png]

Measured accuracy across the recorded trials

Quantity	Result
Trials recorded	5
Target column correctly identified	5/5
Book row identified	5/5
Book 3D position measured	5/5
Mean |offset| of book from column centre	0.145 m
Largest |offset| handled	0.254 m

Denominator is the number of recorded trials. A trial counts as identified only if the value was published from an image, never from the launch argument.

3 Navigation

The base is a mecanum holonomic drive, so x, y and yaw are independent. Rafeeq exploits that twice: it strafes along the shelf to put a jittered book squarely in front of the arm without turning away from the shelf, and it performs the entire final grasp approach as a pure base translation while the arm holds one validated posture.

Control is proportional on the odometry error, rotated into the body frame, with position tolerance 0.02 m and yaw tolerance 0.02 rad. Obstacle checking uses both 270-degree lidars binned into 36 sectors: the commanded direction is checked against the returns that actually lie in it, which a front-only test cannot do on a robot that can strafe and reverse. Stale odometry or scans stop the base rather than letting it integrate against a frozen pose. "Blocked" is reported separately from "reached": an earlier version conflated them and could believe it had arrived at a grasp pose it had never reached.

Travel time

The simulator ran at a measured real-time factor of 0.362 on the validation machine, so simulated and wall-clock seconds differ by nearly threefold. Both are recorded; the table below is simulated time, which is the machine-independent figure.

4 Manipulation

Grasping is where nearly all of the engineering went, and where the remaining blocker is. Four measured facts fix the problem almost completely.

- A book presents its spine, 0.03 m wide. Only that dimension fits between the pads, so the fingers must close along world y. This leaves the gripper roll as the single free parameter.
- The gripper body is 0.157 m long along the axis perpendicular to the fingers. With a horizontal approach that axis is vertical - and a shelf row is 0.300 m tall with a 0.250 m book already in it. The body does not fit.
- Tilting the approach rolls that body axis towards the horizontal, out of the row. Measured vertical extent of the gripper body: 0.174 m at 0 deg, 0.123 m at 45 deg, 0.060 m at 70 deg, 0.000 m at 90 deg. Whole-arm shelf intrusion reaches zero from 45 deg onwards.
- The pads themselves separate as gap = 0.0051 + 0.777 x joint, measured by transforming the fingertip collision mesh into the grasp frame. The fingertip link origins give a different and misleading curve.

The sequence

```
observe -> stage 1.35 m back -> plan (roll x height x tilt) -> take posture -> base translates in   -> refine against
live base pose -> close -> verify contact -> lift -> base retreats -> verify retention
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The arm takes its posture while still 1.35 m from the shelf, where it physically cannot reach the shelf while swinging. The base then carries it in. Re-solving IK at each step of an arm-driven approach changes the arm posture inside a 0.30 m gap on every step, and the controller interpolates straight through whatever lies between two postures.

Verification, and why it is not the joint position

gz_ros2_control implements a position command interface as a JointVelocityCmd computed from position error, so the finger joint is driven kinematically and reaches whatever it is commanded to reach. Measured, commanded and achieved positions agree to five decimals whether or not a book is between the pads, and the reported effort is a constant -0.0935 N. A stall therefore cannot be used as evidence of a grasp. Rafeeq instead reads the fingertip contact sensors and requires the contact to persist through the lift and through a 0.42 m base retreat.

5 Results

The delivering run

The task has been completed end to end on the live simulator, in one continuous unedited take, and the completion was reported by the organisers' own contact sensor rather than by our gripper opening. Seed 101, shelf column 2, red book. The line that ends the run is:

```
OFFICIAL COMPLETION EVENT: book_col_3_row_2_red contacted the collection bin at t+107.8 s
```

The model name is the simulator's own, read back afterwards as a consistency check and never used to choose anything: the plate numbered 2 hangs above physical slot 3, and SDF row_2 is competition row 1. The run's own evidence image records the mapping it read - "requested column 2 to physical shelf slot 3", plates 3@slot1, 5@slot2, 2@slot3, 1@slot4, 4@slot5.

Quantity	Value	Where it comes from
Delivery	t+107.8 simulated s	organiser /bin_contacts
Trial timer / video	221.3 wall s · 4:34 take	realised real-time factor 0.487
Carry, grasp to release	7.8 simulated s	against a grip good for 6-11 s
Collisions	1 (arm_right_6_link, during RETREAT)	organiser /contacts

Two clocks, and they are not the same clock: the mission counts simulated seconds (use_sim_time), the video timer counts wall seconds from solution.launch.py. The take has no machine-readable record - the recorder ends on the delivery event, before the run writes one - so its logs, frames and checksums are committed at media/ and it is excluded from every rate below.

Recorded trials

These are the machine-readable trial records in results/, one JSON per run, written by the solution itself. They predate the extraction work described in section 4 and every one of them ends at the retention check or earlier, so they are the identification and measurement evidence, not delivery evidence. They are kept rather than replaced because that is what they honestly show.

run	req col/col	slot	row	offset m	grasp	deliv	coll	score	sim s
T01_seed101_col2_red	2 / red	3	1	-0.151	no	no	0	6.0	83
T02_seed202_col4_blue	4 / blue	2	4	-0.009	no	no	0	6.0	104
T03_seed303_col1_yello	1 / yellow	4	1	-0.132	no	no	0	6.0	83
T04_seed404_col5_green	5 / green	2	2	-0.254	no	no	0	6.0	121
T05_seed505_col3_red	3 / red	4	3	+0.179	no	no	1	5.5	104

Score is a LOCAL ESTIMATE from the implemented rubric, not an official result: gentle-versus-drop and collision intent are judge decisions. Times are simulated seconds. These runs were recorded in software rendering at a real-time factor of 0.362; on the GPU the simulator runs at 0.65-0.70 headless and the delivering take realised 0.487 with the viewer up.

Rates

Metric	Value	Denominator
Column identified correctly	5/5	recorded trials
Book row identified	5/5	recorded trials
Book position measured	5/5	recorded trials
Grasp verified	0/5	recorded trials
Delivered to bin	0/5	recorded trials (all predate the fix)
Collision events (total)	1	across all trials
Mean estimated score	5.9 / 19	recorded trials
Mean duration	99 sim s	recorded trials

All 5 reached the shelf and closed on the right book. 4 then failed at retention - the fingertips lost the book after the lift or during extraction - and 1 was an IK refusal at the refine step. That is the problem of section 6, and the work since (the above-centre pinch, the ramped retreat, the direct carry) is what turned it into the delivering run above. It costs more than it looks: the rubric awards shelf navigation on a successful grasp and the return on a successful delivery, so a failure at the grasp costs 13 of the 19 points, not 3. Two further trials exceeded the harness wall-clock cap before reaching a terminal state and wrote no record; they are excluded rather than counted either way.

6 Limitations and future work

The grip is a clock

Grasp retention was the open blocker of the previous revision. It no longer blocks - the book is grasped, extracted, carried and delivered - but it is not solved, and the distinction matters. The pads sit on mimic joints whose URDF effort limit is 0.1 N m, gz_ros2_control enforces a mimic with a velocity servo capped at that limit, and nothing in the provided world offers an attach mechanism. The grip is therefore friction against a hinge that can be pushed open: the book creeps out of the pads under its own weight and stays held, measured, for 6 to 11 seconds once it hangs free. Squeezing harder does not help - 6 mm to 16 mm of interference changed nothing, and past a point it simply loads those hinges.

What made delivery possible was accepting that and sizing the task to it: the bin is identified before the shelf trip, so no time is spent hunting for it with a book in hand; the drive starts while the hold is still being confirmed; and the arm holds still throughout the carry, because every arm motion swings the book. The delivering take carried it for 7.8 s against a grip good for 6 to 11. That margin is why runs still fail - of two validation trials after the fix, one dropped the book 3.0 s into the carry and one lost it during extraction, both having already identified, measured and grasped the correct book. Widening the margin needs a higher effort limit on those joints or an attach mechanism in the world: properties of the provided environment, not of this approach. The measurements, the reproducer and the four remedies tried and reverted are in docs/grasp_blocker.md.

What changed during this work

Defect found	Resolution
16 states treated a timeout as success	Every transition now gates on an observation
Measured book position published, never consumed	Grasp planner consumes it; base strafes to it
Grasp poses aimed at column centres	Books jitter 0.25 m; poses solved per measurement
Top row unreachable	Wrong wrist roll; both rolls now tried
Arm drove through shelf boards	Torso set from row height; roll/tilt search
Gripper body fouled the row	Approach tilt search, 0 shelf contacts
Bin pose used the spawn height (1.3 m)	Measured settled geometry, rim at 0.950 m
Both grippers parked inside the shelf	Genuine tucks for both arms
Trajectories timed at one guessed speed	Per-joint URDF velocity limits
Evidence images excluded by .gitignore	Now committed as required evidence

Remaining work

- Widen the retention margin rather than living inside it, then re-run the trial matrix against the delivering solution: one delivery is not a measured rate.
- Replace the coarse advisory clearance model - a padded polyline through link origins - with proper swept-volume checking. It is good enough to rank IK solutions and not good enough to veto them.
- Eliminate the start-up collision in which the tucked left arm brushes the table as the base turns to face the shelf.
- Cover all 20 column/colour combinations and several randomised layouts, rather than the recorded subset.

Simulation versus physical deployment

Nothing here has run on hardware, and the central finding of this report is specific to the simulator: a kinematic gripper on a 0.1 N m mimic hinge behaves nothing like a real TIAGo Pro applying a real force to a continuous contact. Conversely, a physical robot brings problems this work has not faced - odometry drift over a long run, lighting that would break tight HSV bands, and books that are not uniform saturated primaries. The perception approach would need re-grounding in real imagery before any of it transfers.

Longer-term intent

Rafeeq is named for the Arabic for companion, and the intent behind it is a library assistant that fetches books for people who cannot reach or navigate the stacks. None of that accessibility work is implemented - no voice interface, no accessibility testing, no deployment with People of Determination. It is the direction of travel, not a claim about this submission.